

- 4 Fig. 4.1 shows a fixed mass of ideal gas in a cylinder of pressure $2.1 \times 10^5 \text{ Pa}$, volume $4.0 \times 10^{-4} \text{ m}^3$ and temperature 27°C .

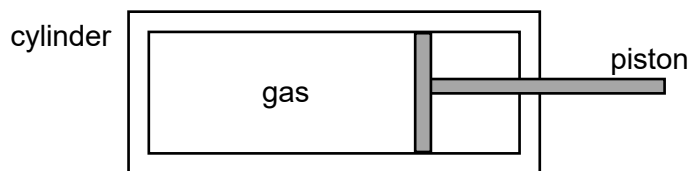


Fig. 4.1

The gas is compressed at constant temperature along process I. Fig. 4.2 shows the variation with volume V of the pressure P of the gas.

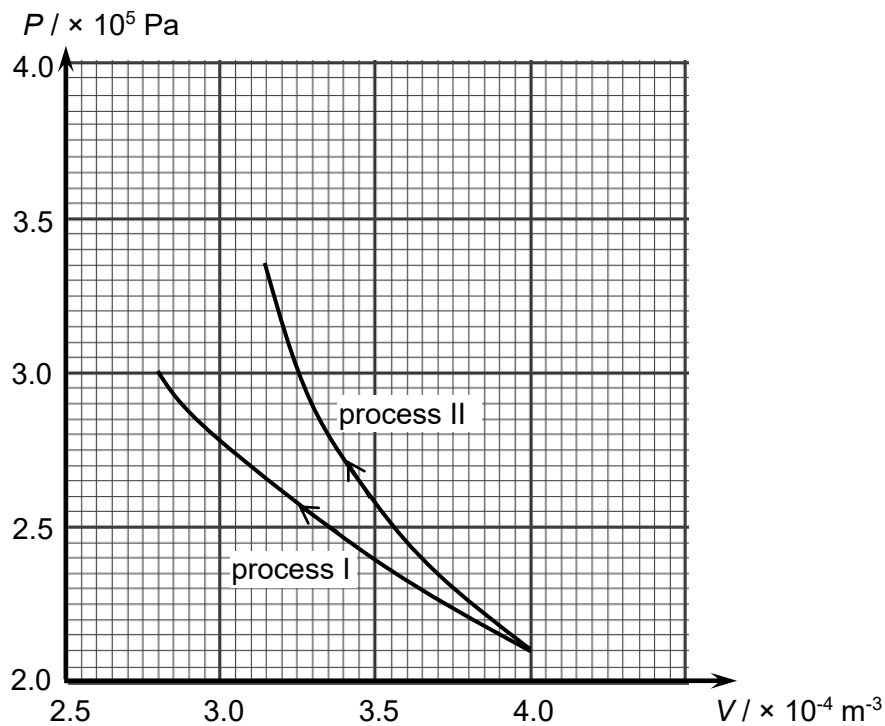


Fig. 4.2

- (a) (i) With reference to Fig. 4.2, estimate the work done on the gas through process I.

$$\begin{aligned}
 \text{Work done on gas} &= \text{Area under } P-V \text{ graph} \\
 &= 0.5(2.90 + 2.10) \times 10^5 \times (4.00 - 2.80) \times 10^{-4} \\
 &= 30.0 \text{ J}
 \end{aligned}$$

work done = J [3]

- (ii) State the *first law of thermodynamics*.

The increase in internal energy is the sum of the heat supplied and the work done on system. [1]

[1]

- (ii) Determine the heat loss from the gas through process I.

30.0 J. Since there is no change in temperature, there is no change in internal energy. Hence heat loss by gas is equal to work done on the gas.

[1]

- (b) A second identical cylinder containing the same ideal gas is thermally insulated. The gas is compressed to a new pressure and volume, as shown in process II. The work done on the gas in process I equals to the work done on the gas in process II.

- (i) Using the kinetic theory of gases, explain why the pressure in process II increases.

There is no heat loss to surrounding, and the work done on gas results in an increase in internal energy. [B1]

This increase in internal energy is equivalent to the increase in kinetic energy, resulting in an increase in speed / momentum of the gas particles. [B1]

Upon collision with the walls of the cylinder, there is a greater change in momentum. [B1]

There is also higher frequency of collision between particles in the walls of the cylinder. [B1]
These result in a greater force and hence greater pressure.

- (ii) Calculate the final temperature of process II.

$$\frac{P_i V_i}{T_i} = \frac{P_f V_f}{T_f}$$

$$\frac{(2.10)(4.00)}{273.15 + 27} = \frac{(3.35)(3.15)}{T_f}$$

$$T_f = 377 \text{ K} = 104 \text{ }^{\circ}\text{C}$$

temperature = $^{\circ}\text{C}$ [2]

[Total: 11]

